DYLAN WILSON

dylan.p.wilson.27@dartmouth.edu • Dartmouth College, Class of 2027 • GitHub • Website • LinkedIn

EDUCATION

Dartmouth College Class of 2027 B.A. in Mathematics & Psychology • GPA: 3.7

RESEARCH

Applying Godel Agents to Idea Generation · Independent (collab. WashU) · 2025–Present

• Combines Developed a self-improving agent framework for scientific hypothesis generation using iterative reasoning and tool-based refinement loops

.• Implemented an evaluation pipeline measuring hypothesis quality, reasoning depth, and improvement across iteration. Achieves SOTA on idea generation (measured by ELO Bradley comparisons).[Github] [Workshop paper]

Verifiable Self-Learning via Dual-Play RL · with Jiaxin Huang · WashU · 2025–Present

• Challenger model generates questions optimized for verifiability; solver attempts answers under judge evaluation.

• Core finding: judge agreement is a reliable proxy for question verifiability—the subset judges consistently agree on drives a correctness improvement from ~60% to ~90%.[Github]

Evolutionary Learning & Signal Stability · with Prof. Daniel Rockmore · Dartmouth College · 2025

• Built evolutionary training workflows including compatibility-based pairing and task-specific LoRA optimization.

• Investigated instability in reinforcement-style learning systems, identifying diversity–signal tradeoffs and collapse under excessive optimization pressure.

• Concluded that training signal reliability, rather than optimization method, is the central bottleneck in scalable model learning. [Github][Github][Paper]

Tree-Search Reasoning System · with Prof. Yu Wing Tai · Dartmouth College · 2024

• Confidence-guided tree search over candidate responses; ~20% BLEU improvement vs. single-pass generation.[Paper]

INDUSTRY EXPERIENCE

Research Engineer · DeepGroove · San Francisco, CA · 2025

• Built an asynchronous data generation system on vLLM to support training of deep-research models using a lightweight BitNet architecture.

• Generated 30,000+ synthetic research queries, enabling large-scale training experiments.

• Implemented rubric-based scoring using Elo-style comparisons for RL without fixed labels.[Write-up]

Software Engineer / Project Lead · Primer.ai · Washington, DC · 2024

• Built automated phone-calling service prototype deployed globally as project lead.

• Developed internal evaluation tooling measuring model performance under real-world conditions.

OTHER TECHNICAL WORK

• LLM jailbreak study (responsible disclosure): 0-shot strategy achieving ~92% success rate across frontier models.[Paper]
•Lang2Logic: Python package with 5,000+ downloads translating natural language into structured Python objects for LLM-logic interaction and self editing agent workflows. [GitHub]

•Built self-improving Gödel-style agent loops (initiated in high school); early precursor to current research direction [GitHub]

• EARLY EXPERIENCE — Caltech Neuroeconomics Lab, SupplyFrame, Virtualitics, Scratch Financial